



HOGWARTS SCHOOL FOR WIZARDS & WITCHES

2020

Practise
Trial HSC
Examination

Mathematics

HSC 2014 Paper updated for new syllabus

This paper breaks just about every copyright law on the planet.

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using a black pen.
- NESA approved calculators may be used.
- A reference sheet is provided.
- Write your candidate number and/or name on each answer booklet and on your multiple-choice answer sheet.
- Place all your booklets inside the answer booklet for Question Eleven.

Total Marks - 100

Section I – 10 Marks

- Questions 1-10 are of equal value.
- Record your answers to the multiple choice on the sheet provided.

Section II – 90 Marks

- Questions 11-16 are of equal value.
- All necessary working should be shown.
- Write your answers with the relevant working out on the answer sheet provided.

Name:

Teacher:

THIS IS A TRIAL PAPER ONLY

It does not necessarily reflect the format or the content of the 2020 HSC Examination Paper in this subject. In fact, it definitely doesn't!

Section I

10 marks

Attempt Questions 1 - 10

Allow about 15 minutes for this section

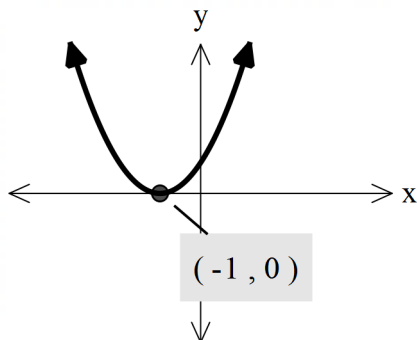
Use the multiple-choice answer sheet for Questions 1-10

1 Experimental results indicate the correlation between the mass of food remaining, m , and the number of ants eating the food, n , is -0.4 . Which of the following statements is true?

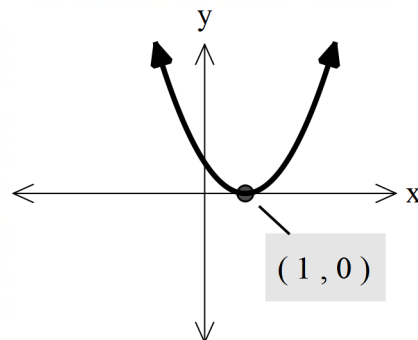
- (A) 40% of the food has been eaten by the ants.
- (B) Increasing the amount of food increases the number of ants.
- (C) Increasing the number of ants decreases the amount of food.
- (D) 60% of the change in number of ants is due to random factors.

2 Which graph best represents $y = (x - 1)^2$?

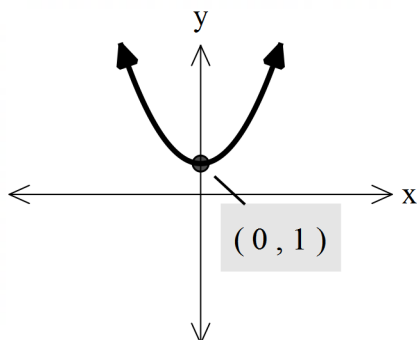
(A)



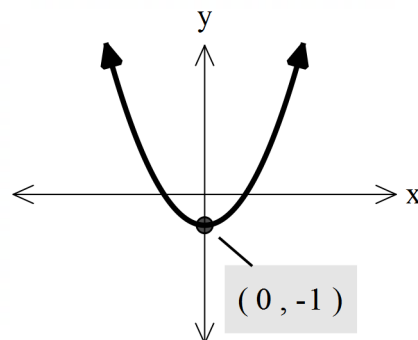
(B)



(C)

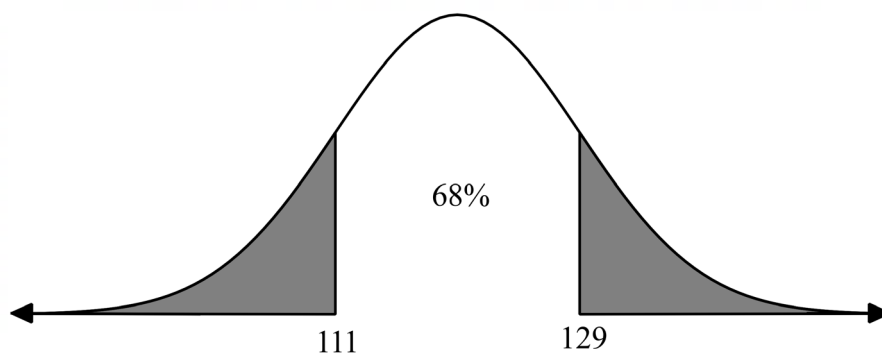


(D)



- 3 What is the solution to the equation $\log_2(x-1) = 8$?
- (A) 4
(B) 17
(C) 65
(D) 257
- 4 Which expression is equal to $\int e^{2x} dx$?
- (A) $e^{2x} + c$
(B) $2e^{2x} + c$
(C) $\frac{e^{2x}}{2} + c$
(D) $\frac{e^{2x+1}}{2x+1} + c$
- 5 Which equation represents the line perpendicular to $2x - 3y = 8$, passing through the point $(2, 0)$?
- (A) $3x + 2y = 4$
(B) $3x + 2y = 6$
(C) $3x - 2y = -4$
(D) $3x - 2y = 6$
- 6 The standard normal distribution is such that $Z \sim N(0,1)$. That is to say, it has a mean of 0 and a variance of 1.

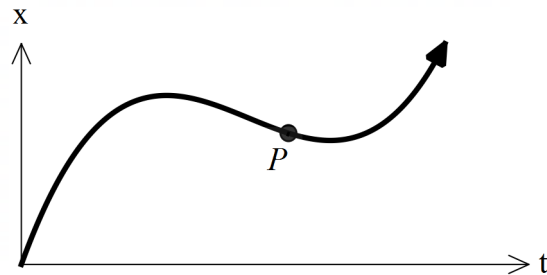
Consider the following graph.



The graph above is best described by:

- (A) $N(111, 129)$
(B) $N(120, 81)$
(C) $N(120, 3)$
(D) $N(120, 9)$

- 7 How many solutions of the equation $(\sin x - 1)(\tan x + 2) = 0$ lie between 0 and 2π ?
- (A) 1
(B) 2
(C) 3
(D) 4
- 8 Which expression is a term of the geometric series $3x - 6x^2 + 12x^3 - \dots$?
- (A) $3072x^{10}$
(B) $-3072x^{10}$
(C) $3072x^{11}$
(D) $-3072x^{11}$
- 9 The graph shows the displacement x of a particle moving along a straight line as a function of time t .



- (A) The velocity is negative and the acceleration is positive.
(B) The velocity is negative and the acceleration is negative.
(C) The velocity is positive and the acceleration is positive.
(D) The velocity is positive and the acceleration is negative.
- 10 Three runners compete in a race. The probabilities that the three runners finish the race in under 10 seconds are $\frac{1}{4}$, $\frac{1}{6}$ and $\frac{2}{5}$ respectively. What is the probability that at least one of the three runners will finish the race in under 10 seconds?
- (A) $\frac{1}{60}$
(B) $\frac{37}{60}$
(C) $\frac{3}{8}$
(D) $\frac{5}{8}$

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours and 45 minutes for this section

Answer each question in the appropriate writing booklet.

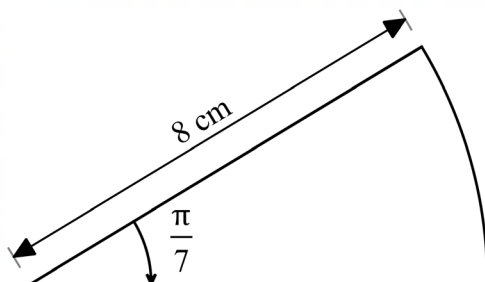
In Questions 11-16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)	Marks
(a) Rationalise the denominator of $\frac{1}{\sqrt{5}-2}$.	2
(b) Let X be a normally distributed random variable with mean 11 and variance 4, and let Z be the random variable with the standard normal distribution. Find $Pr(X < 7)$.	2
(c) Differentiate $\frac{x^3}{x+1}$.	2
(d) Find $\int \frac{1}{(x+3)^2} dx$.	2
(e) Evaluate $\int_0^{\frac{\pi}{2}} \sin \frac{x}{2} dx$.	3
(f) The gradient function of a curve $y = f(x)$ is given by $f'(x) = 4x - 5$. The curve passes through the point (2, 3). Find the equation of the curve.	2

Question 11 (continued)

- (g) The angle of a sector in a circle of radius 8 cm is $\frac{\pi}{7}$ radians, as shown in the diagram.

2



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Find the exact value of the perimeter of the sector.

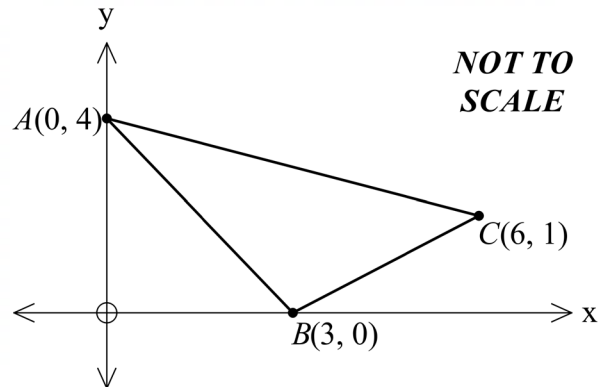
Question 12 (15 marks)

Marks

- (a) Evaluate the arithmetic series $2 + 5 + 8 + 11 + \dots + 1094$.

2

- (b) The points $A(0, 4)$, $B(3, 0)$ and $C(6, 1)$ form a triangle, as shown in the diagram.

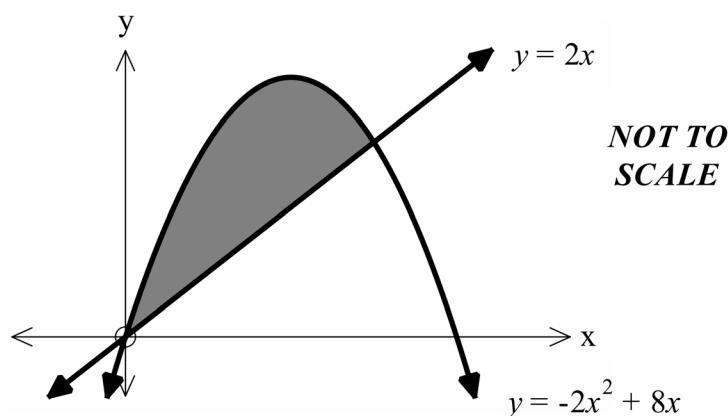


- (i) Show that the equation of AC is $x + 2y - 8 = 0$. 2
- (ii) Find the point at which AC would intersect with the x -axis if it were extended. 2
- (iii) Hence, or otherwise, find the area of $\triangle ABC$. 2

- (c) A packet of lollies contains 5 red lollies and 14 green lollies. Two lollies are selected at random without replacement.

- (i) Draw a tree diagram to show the possible outcomes. Include the probability on each branch. 2
- (ii) What is the probability that the two lollies are of different colours? 1

- (d) The parabola $y = -2x^2 + 8x$ and the line $y = 2x$ intersect at the origin and at the point A.



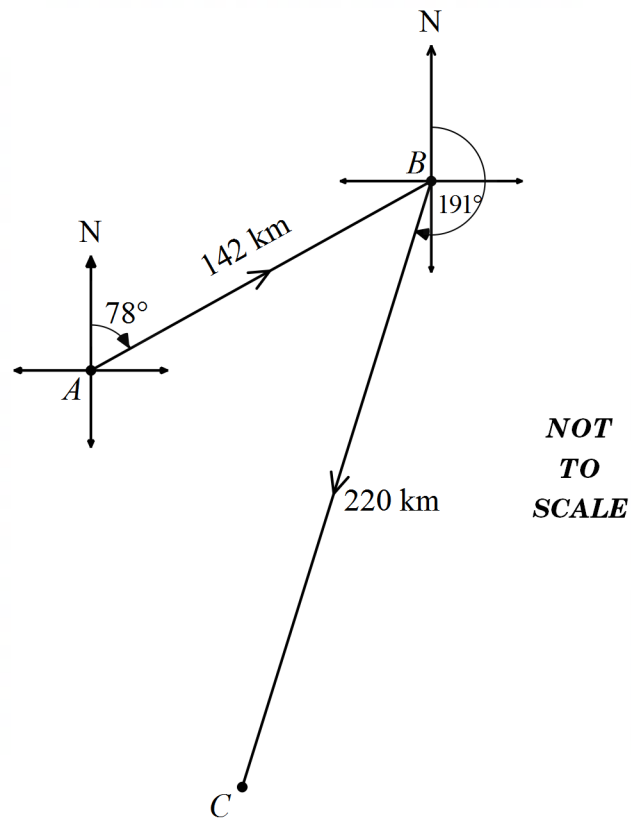
- (i) Find the x -coordinate of the point A. 1
- (ii) Calculate the area enclosed by the parabola and the line. 3

Question 13 (15 marks)**Marks**

- (a) (i) Differentiate $3 + \sin 2x$ 1
- (ii) Hence, or otherwise, find $\int \frac{\cos 2x}{3 + \sin 2x} dx$. 2
- (b) Two events, A and B are independent, where $P(B) = 2P(A)$ and $P(A \cup B) = \frac{5}{8}$. What is the value of $P(A)$? 3
- (c) The population P of a certain insect is falling according to the formula $P = 3000e^{-kt}$, where t is in days. 1
- Show that $\frac{dP}{dt} = -kP$.
- (d) The displacement of a particle moving along the x-axis is given by
- $$x = t - \frac{1}{1+t},$$
- where x is the displacement from the origin in metres, t is the time in seconds, and $t \geq 0$.
- (i) Show that the acceleration of the particle is always negative. 2
- (ii) What value does the velocity approach as t increases indefinitely? 1

Question 13 (continued)

- (e) Chris leaves island A in a boat and sails 142 km on a bearing of 078° to island B. Chris then sails on a bearing of 191° for 220 km to island C, as shown in the diagram.

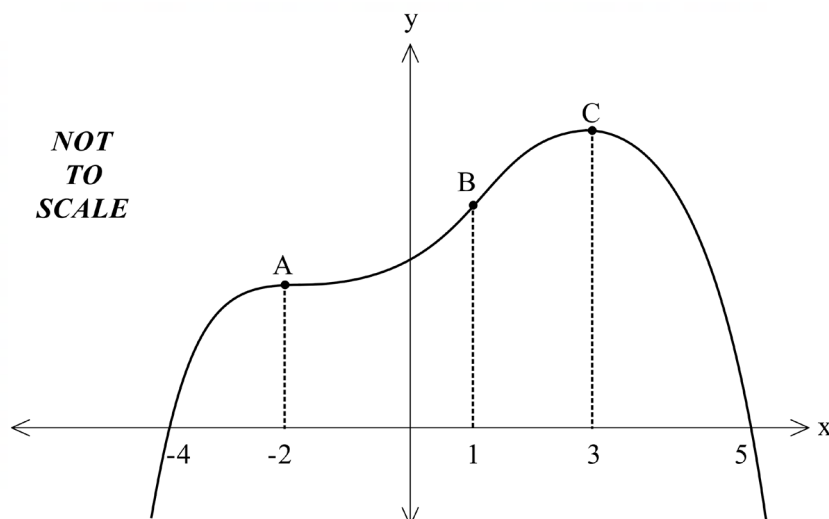


- | | |
|---|---|
| (i) Show that the distance from island C to island A is approximately 210km. | 2 |
| (ii) Chris wants to sail from island C directly to island A. On what bearing should Chris sail? Give your answer correct to the nearest degree. | 3 |

Question 14 (15 marks)

Marks

- (a) Find the coordinates of the stationary point on the graph $y = e^x - ex$, and determine its nature. 3
- (b) Sketch the curve $y = 3 + 2 \sin 2x$ for $0 \leq x \leq 2\pi$. 3
- (c) Solve $2 \sin^2 \frac{x}{3} = 1$ for $-\pi \leq x \leq \pi$. 3
- (d) At the beginning of every 8-hour period, a patient is given 10 mL of a particular drug. During each of these 8-hour periods, the patient's body partially breaks down the drug. Only $\frac{1}{3}$ of the total amount of the drug present in the patient's body at the beginning of each 8-hour period remains at the end of that period.
- (i) How much of the drug is in the patient's body immediately after the second dose is given? 1
- (ii) Show that the total amount of the drug in the patient's body never exceeds 15 mL. 2
- (e) The diagram shows the graph of a function $f(x)$. 3
- The graph has a horizontal point of inflexion at A , a point of inflexion at B and a maximum turning point at C .

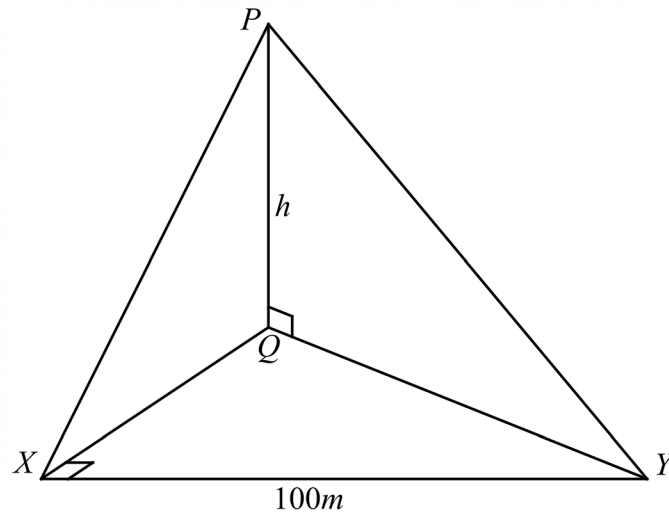


Sketch the graph of the derivative $f'(x)$.

Question 15 (15 marks)

Marks

- (a) Find all solutions of $2\sin^2 x + \cos x - 2 = 0$, where $0 \leq x \leq 2\pi$. 3
- (b) The angle of elevation of a tower, PQ , from a point X due south of it is observed to be 70° . From another point Y , due east of X , the angle of elevation is 58° .



Given $XY = 100m$, find the height h of the tower PQ to the nearest metre.

- (c) Consider the curve $y = xe^{-x}$.
- (i) Find any x or y -intercepts. 1
 - (ii) Find any stationary points, and determine their nature. 1
 - (iii) Use your calculator, or otherwise, to determine the behaviour of the curve as $x \rightarrow \pm\infty$. 1
 - (iv) Hence, sketch the curve. 1
- (d) The line $y = mx$ is a tangent to the curve $y = e^{2x}$ at a point P .
- (i) Sketch the line and the curve on one diagram. 1
 - (ii) Find the coordinates of P . 3
 - (iii) Find the value of m . 1

Question 16 (15 marks)**Marks**

- (a) Use the Trapezoidal Rule with five function values to show that

3

$$\int_{-\frac{\pi}{3}}^{\frac{\pi}{3}} \sec x \, dx \approx \frac{\pi}{9} \left(3 + \frac{8}{\sqrt{3}} \right).$$

- (b) At the start of a month, Jo opens a bank account and makes a deposit of \$500. At the start of each subsequent month, Jo makes a deposit which is 1% more than the previous deposit.

At the end of every month, the bank pays interest of 0.3% (per month) on the balance of the account.

- (i) Explain why the balance of the account at the end of the second month is

2

$$\$500(1.003)^2 + \$500(1.01)(1.003).$$

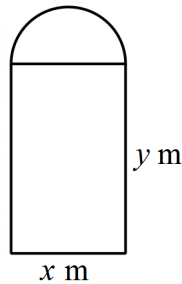
- (ii) Find the balance of the account at the end of the 60th month, correct to the nearest dollar.

3

Question 16 (continued)

- (c) The diagram shows a window consisting of two sections. The top section is a semicircle of diameter x m. The bottom section is a rectangle of width x m and height y m.

The entire frame of the window, including the piece that separates two sections, is made using 10 m of thin metal.



The semicircular section is made of coloured glass and the rectangular section is made of clear glass.

Under test conditions the amount of light coming through one square metre of the coloured glass is 1 unit and the amount of light coming through one square metre of the clear glass is 3 units.

The total amount of light coming through the window under test conditions is L units.

- (i) Show that $y = 5 - x\left(1 + \frac{\pi}{4}\right)$. 2
- (ii) Show that $L = 15x - x^2\left(3 + \frac{5\pi}{8}\right)$. 2
- (iii) Find the values of x and y that maximise the amount of light coming through the window under test conditions. 3